

Qatar-4b

R-2507

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_211122 · created 2026-08-24T23:52:57+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459540.6252 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.054
Transit depth (Rp/Rs)^2	3.38 +/- 1.99 [%]
Orbital Inclination (inc)	88.73 +/- 2.12
Airmass coefficient 1 (a1)	2.53 +/- 0.61
Airmass coefficient 2 (a2)	-0.88 +/- 1.16
Scatter in the residuals of the lightcurve fit is	24.57 %
Best Comparison Star	#2 - [241, 214]
Optimal Aperture	1.27
Optimal Annulus	8.48
Transit Duration (day)	0.093 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.054 vs 0.138 (deviation 0.85σ)
Duration fit vs published	0.093 d vs 0.0892 d (deviation 0.6σ)
Mid-time O–C	-5.2 min
Depth SNR	3.4
Residual scatter	24.57 %

Light curve

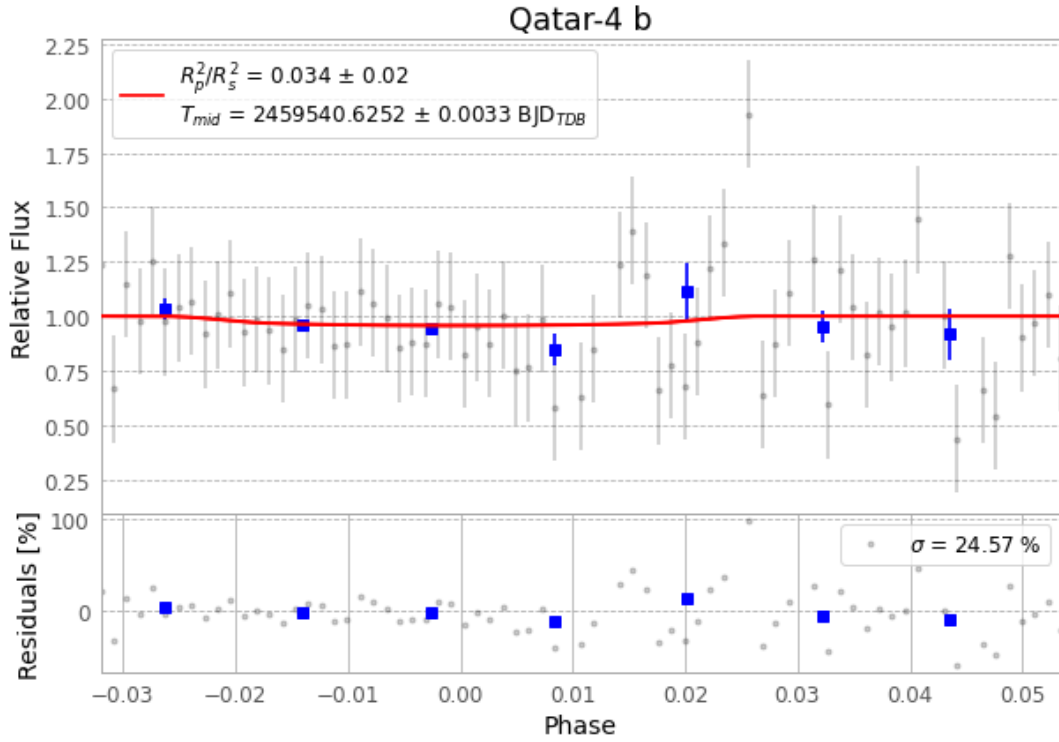


Figure 1 — FinalLightCurve_Qatar-4 b_2021-11-21.png (SHA-256 df722a12c838d1d0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 79f4d331d7bbf36b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), Qatar-4211122013013.FITS → Qatar-4211122051212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-211122.log (2c64f57f1731...), FinalLightCurve_Qatar-4 b_2021-11-21.png (df722a12c838...), FinalLightCurve_Qatar-4 b_2021-11-21.csv (053988a2a578...)

Compute resources

Wall time 138.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:52Z.